

Risk Measures

Objective: to present the tools of quantitative management of financial risks, their empirical motivations, theoretical properties and statistical inference.

References:

- McNeil, A.J., Frey, R. and P. Embrechts (2005) Quantitative risk Management, Princeton University Press.
- Pamplemousse
- My [web page](#)

Plan:

- Chapter 1: Basic concepts (VaR, ES, ...)
- Chapter 2: Coherent risk measures
- Chapter 3: Statistical Inference

Risk measures

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Chapter 1: Basic concepts for quantitative analysis of financial risks

Measuring risks in Finance/Insurance

Investors, banks and others **financial market players** have their capital exposed to *risk*. It is important to quantify the risk of a particular position to decide whether it is acceptable or not.

Following the recent financial crises which too frequently resulted in an undervaluation of reserves, new **regulation** measures were taken: for financial institutions (Basel II-III), and insurance companies (Solvency II). Regulators encourage banks and insurance companies to develop their **own risk assessment systems**.

For these reasons, **many risk assessment methods** have been developed.

Plan

- 1 Financial Risks
 - Various types of risks
 - Regulation
 - Risk factors
- 2 Main risk measures
- 3 Examples of methods to assess the risk of a portfolio

Major categories of risks

Investors, banks and other financial institutions, as well as insurance companies, are exposed to financial risks:

- ① **credit risk** (default of a borrower)
- ② **market risk** (fall in the value of an asset, exchange rate, commodity prices, interest rate risk, ...)
- ③ **operational risk** (IT system failure, a Covid or flu epidemic, fire, internal fraud, external fraud, legal risk, commercial disputes, disputes with the authorities, ...)
- ④ **liquidity risk** Market liquidity risk (not being able to sell or buy quickly enough, no counterparty, excessive transaction volume leading to price fluctuations) and funding liquidity risk (failure to fulfill withdrawal requests)

Major categories of risks

There is also:

- **counterparty risk** (for example on a credit default swap (CDS))
- **model risk** (using a wrong model to measure risk)
- **systemic risk** (collapse of the entire financial system caused by the uniformity of asset portfolios, the interdependence of banks, or the default of a Systemically Important Financial Institution (SIFI)...))
- **contagion risk** (increase in asset correlations)
- **concentration risk** (all eggs in one basket)
- **moral hazard risk** (a bank which is considered as "too big to fail" has no incentive to be cautious)
- **natural disaster risk**, **demographic risk**, ... some can certainly be integrated into operational risk.

Basel Committee

The Basel Committee on Banking Supervision was established by the governors of the G10 central banks in 1985. The committee makes recommendations (which do not have the force of law) on banking prudential behavior:

- ① **Basel 1** (88): Essentially concerns credit risk. For banks, implementation of a minimum "Cooke ratio" of 8 % equity with respect to the loans granted (with weightings, Risk Weighted Assets (RWA), possibly the weight 0 for German Sovereign loans and the weight 0.5 for Argentina).
- ② **Basel 2** (04):
- ③ **Basel 3** (published in 2010, implemented in 2013-2019):

Basel Committee

- ① **Basel 1** (88):
- ② **Basel 2** (04): Based on 3 pillars: Capital requirement; Monitoring by regulators (ACPR*, AMF†); Transparency. Also takes into account market risk and operational risk. Use of VaR (since 1993) and other risk measures. Financial institutions can set up their **own risk management model**. The model is evaluated by the regulator, which imposes new capital if the model is not satisfactory.
- ③ **Basel 3** (13):

*Autorité de contrôle prudentiel et de résolution

†Autorité des marchés financiers

Basel Committee

- ① Basel 1 (88):
- ② Basel 2 (04):
- ③ Basel 3 (13):
 - Increase in equity to 10.5% of risk-weighted assets. Specific requirements for systemic banks.
 - An additional capital cushion can temporarily be imposed by a state to avoid bubbles and procyclicality.
 - Introduction of a leverage ratio (not equivalent to the capital constraint because of the risk weighting, which remains assessed by each bank). A cap of around 33 times equity is imposed (even to assumed low risk assets, such as government bonds).
 - Introduction of liquidity ratios (obligation to hold a sufficient proportion of liquid assets).
 - Introduction of the interest rate risk.

Standard/advanced approaches

Financial institutions have the possibility of developing their own risk management model. Two types of approaches are used:

- The **standard approach** is based on simple and standardized methods (use of marginal distributions, nonparametric static methods, sums of RWA without considering correlations).
- The **advanced approach** is based on relatively sophisticated risk assessment methods (use of conditional distributions, correlations between assets, dynamic models).

The supervisor validates the model or not, and can impose additional reserves based on the quality of the model.

Conditional or marginal loss distribution

Let V_t be the value of a portfolio of risky assets at time t .

- At the horizon h , the **loss** is $L_{t,t+h} = -(V_{t+h} - V_t)$.
- At time t , the future loss $L_{t,t+h}$ is **random**. The distribution of $-L_{t,t+h}$ is called the profit and loss (P&L).

For this random loss, we have:

- the conditional distribution, that is the law of $L_{t,t+h}$ given the information available at time t ;
- the unconditional (or marginal) distribution, that is the law of $L_{t,t+h}$ given only V_t .

Risk factors

The value V_t is function of a vector $\mathbf{Z}_t = (Z_{t,1}, \dots, Z_{t,d})'$ of d risk factors: $V_t = f(t, \mathbf{Z}_t)$.

Example

A European investor invests in the CAC40, the SP500 and the FTSE100. We have

$V_t = a_t \text{CAC}_t + b_t \text{SP}_t \times \text{US}_t + c_t \text{FTSE}_t \times \text{GBP}_t = f(t, \mathbf{Z}_t)$ with
 $\mathbf{Z}_t = (\log \text{CAC}_t, \log \text{SP}_t, \log \text{FTSE}_t, \log \text{US}_t, \log \text{GBP}_t)'$.

Denoting by $\mathbf{X}_t = \mathbf{Z}_t - \mathbf{Z}_{t-1}$ the vector of the increments of the risk factors, the distribution of

$$L_{t,t+1} = f(t, \mathbf{Z}_t) - f(t+1, \mathbf{Z}_t + \mathbf{X}_{t+1}),$$

is determined by that of $\mathbf{X}_{t+1} = (X_{t+1,1}, \dots, X_{t+1,d})'$.

Linear approximation of the loss distribution

We have

$$L_{t,t+1} = f(t, \mathbf{Z}_t) - f(t+1, \mathbf{Z}_t + \mathbf{X}_{t+1}) = \ell_t(\mathbf{X}_{t+1})$$

where $\ell_t(\cdot)$ is a *loss operator*, linking the risk factor increments to the loss.

If f is differentiable, the loss can be approximated by

$$L_{t,t+1} \simeq - \left(\frac{\partial f}{\partial t}(t, \mathbf{Z}_t) + \sum_{i=1}^d \frac{\partial f}{\partial z_i}(t, \mathbf{Z}_t) X_{t+1,i} \right)$$

and the loss operator by

$$\ell_t(\mathbf{x}) \simeq - \left(\frac{\partial f}{\partial t}(t, \mathbf{Z}_t) + \sum_{i=1}^d \frac{\partial f}{\partial z_i}(t, \mathbf{Z}_t) x_i \right)$$

Obviously, the closer f is to linearity and the smaller $\mathbf{X}_{t+1}$, the better the approximation.

Illustration

Example (continued)

For the European investor that invests 30% in CAC40, 40% in SP500 and 30% in FTSE100,

$$V_t = a_1 \text{CAC}_t + a_2 \text{SP}_t \times \text{US}_t + a_3 \text{FTSE}_t \times \text{GBP}_t = 0.3 + 0.4 + 0.3,$$

normalizing such that $V_t = 1$ Euro, we have

$$\begin{aligned} L_{t,t+1} &= 1 - \left(a_1 \text{CAC}_t e^{\log \frac{\text{CAC}_{t+1}}{\text{CAC}_t}} + 0.4 e^{X_{t+1,2} + X_{t+1,4}} + 0.3 e^{X_{t+1,3} + X_{t+1,5}} \right) \\ &\simeq -0.3 X_{t+1,1} - 0.4 (X_{t+1,2} + X_{t+1,4}) - 0.3 (X_{t+1,3} + X_{t+1,5}). \end{aligned}$$

- 1 Financial Risks
- 2 Main risk measures
 - Value at Risk
 - Quantile as solution of an optimization
 - Expected Shortfall
- 3 Examples of methods to assess the risk of a portfolio

Definition

The risk distribution is summarized by the smallest value such that the loss exceeds this value with a probability not exceeding α .

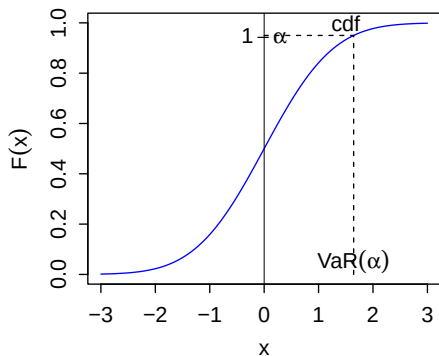
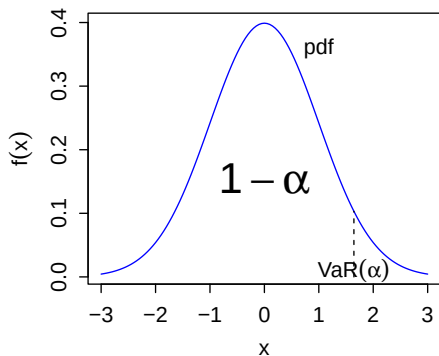
Definition (VaR)

The Value-at-Risk (conditional/marginal), denoted by VaR, at the risk level $\alpha \in (0, 1)$ and at the horizon $h \in \mathbb{N}^*$ is the $(1 - \alpha)$ -quantile of the (conditional/marginal) distribution of the losses:

$$\text{VaR}_{t,h}(\alpha) = \inf \{x : P_t(L_{t,t+h} \leq x) \geq 1 - \alpha\}.$$

If P_t takes into account all the information available up to date t , then $\text{VaR}_{t,h}(\alpha)$ is a conditional VaR. Note that $\text{VaR}_{t,h}(\alpha)$ decreases with α . The reserves needed to cover a position increase with its VaR.

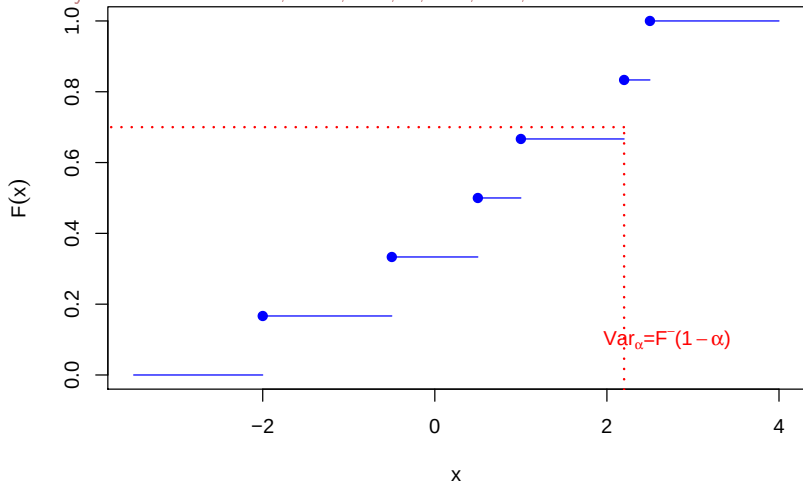
When $F \sim L_{t,t+h}$ is continuous strictly increasing,
 $\text{VaR}(\alpha) := \text{VaR}_{t,h}(\alpha) = F^{-1}(1 - \alpha)$



If F^- is the generalized (pseudo) inverse of F ,

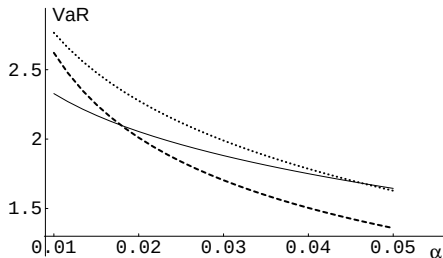
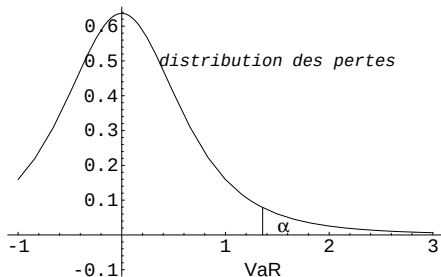
$$\text{VaR}(\alpha) = F^-(1 - \alpha) := \inf\{x : F(x) \geq 1 - \alpha\}$$

X uniformly distributed on $-2, -0.5, 0.5, 1, 2.2, 2.5$; $\alpha = 0.3$



VaR and tail distributions: Gaussian (full line), double exponential (dotted line) and Student with 3 degree of freedom (dashed line)

tails proportional to $e^{-x^2/2}$, $e^{-\sqrt{2}|x|}$ and x^{-4}

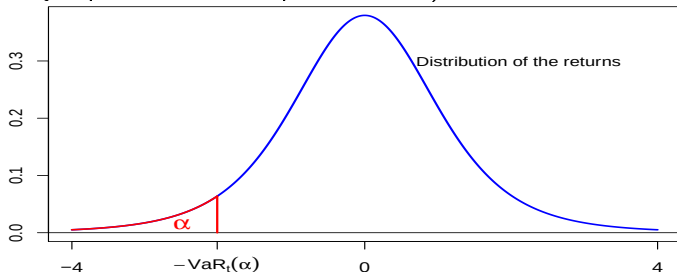


Defining VaR as a return quantile

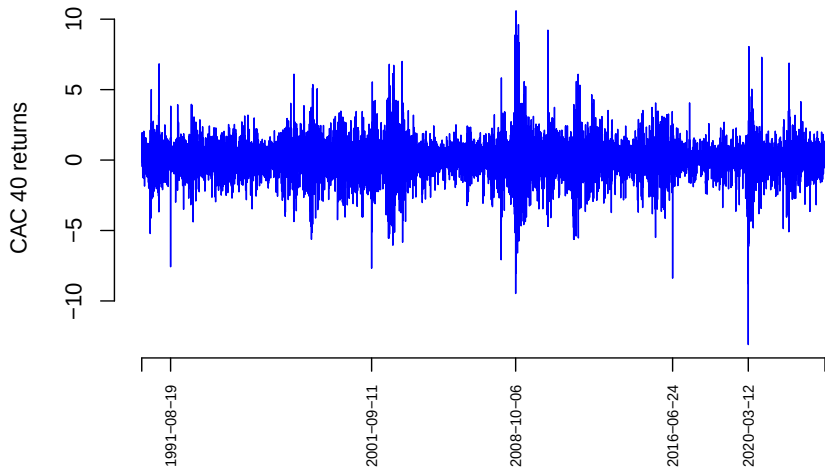
Let the log-returns $r_t = \log V_t / V_{t-1}$, $V_t > 0$, and $q_t(h, \alpha)$ the α quantile of $\sum_{i=1}^h r_{t+i}$. For continuous strictly increasing distributions we have

$$L_{t,t+h} = V_t \left(1 - e^{\sum_{i=1}^h r_{t+i}} \right), \quad \text{VaR}_{t,h}(\alpha) = V_t \left(1 - e^{q_t(h, \alpha)} \right).$$

Sometimes, VaR is defined by $\text{VaR}_{t,1}(\alpha) = -q_t(1, \alpha)$ (new definition, not strictly equivalent to the previous one).



An example of financial returns



The benchmark model for discrete time returns

Daily or weekly financial returns $\epsilon_t = \log(p_t/p_{t-1})$ resemble weak white noise:

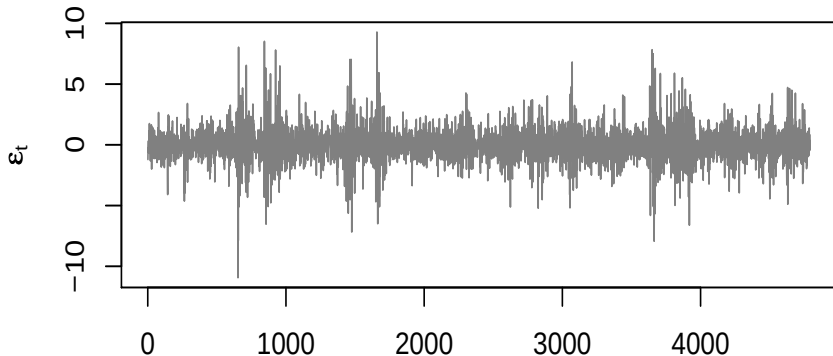
- they are **uncorrelated** (zero autocorrelations), even possibly unpredictable (Fama's efficient-market hypothesis);
- but they are not independent, due to the presence of **volatility clusters** and strong positive autocorrelations of the squares.

Engle (1982) and Bollerslev (1996) proposed the Generalized AutoRegressive Conditionally Heteroscedastic (GARCH) model

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

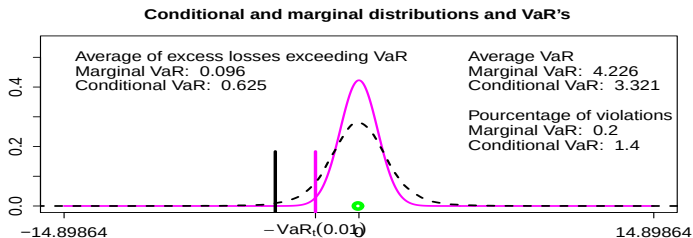
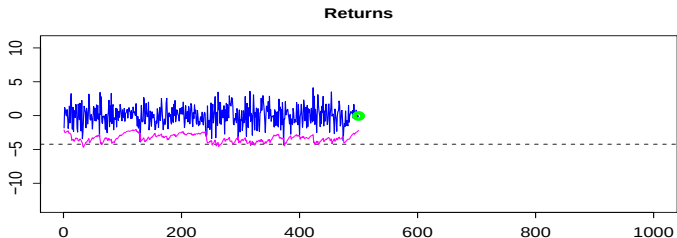
where (η_t) iid $(0,1)$ independent of the past $\mathcal{F}_{t-1} = \sigma(\epsilon_u, u < t)$, $\sigma_t^2 = \text{Var}(\epsilon_t | \mathcal{F}_{t-1})$, with scale $\omega > 0$, reactivity $\alpha > 0$, persistence $\beta \geq 0$, (ϵ_t) strictly stationary ($\alpha + \beta < 1$ sufficient condition).

A simulated trajectory of a GARCH(1,1) model

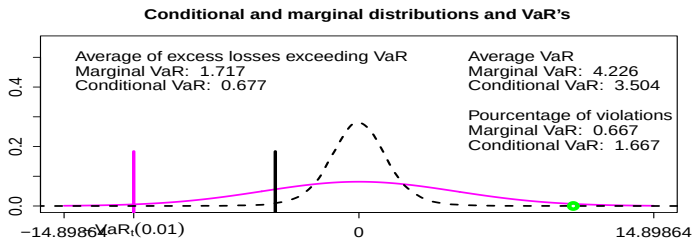
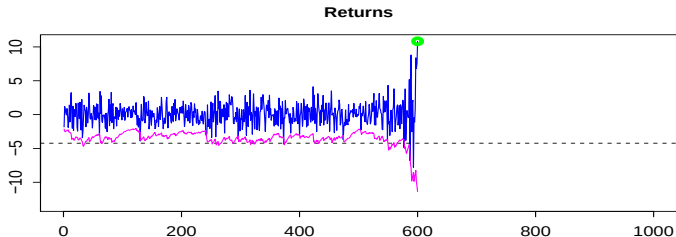


$$\epsilon_t = \sigma_t \eta_t, \eta_t \text{ iid } St_5, \sigma_t^2 = 0.033 + 0.090\epsilon_{t-1}^2 + 0.893\sigma_{t-1}^2, \\ t = 1, \dots, n = 4791$$

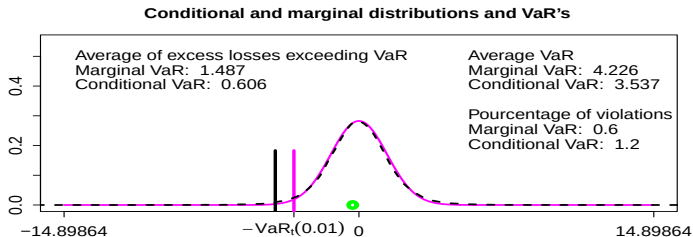
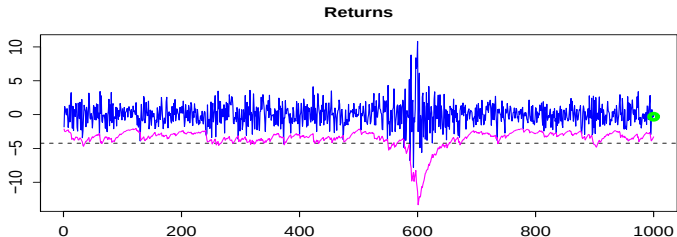
Marginal and conditional VaR on simulated returns



Marginal and conditional VaR on simulated returns



Marginal and conditional VaR on simulated returns



Cumulative distribution function (cdf)

The α -level VaR is the $(1 - \alpha)$ -quantile of the loss distribution. The quantile function is the generalized inverse of the cdf.

Definition (cdf)

The cumulative distribution function (cdf) of a random variable X is the function

$$\begin{aligned} F &: \mathbb{R} \rightarrow [0, 1] \\ x &\mapsto F(x) = P(X \leq x) \end{aligned}$$

A cdf is increasing, tends to 0 at $-\infty$, tends to 1 at $+\infty$, is right continuous (if $x_n \downarrow x$ then $P_X((-\infty, x_n]) \downarrow P_X((-\infty, x])$), has left limits (if $x_n \uparrow x$ then $P_X((-\infty, x_n]) \uparrow P_X((-\infty, x))$) and is continuous in x iff $P(X = x) = 0$.

Quantile function

Definition (Quantile)

The generalized inverse, also called quantile function, of the random variable X with cdf F is defined by

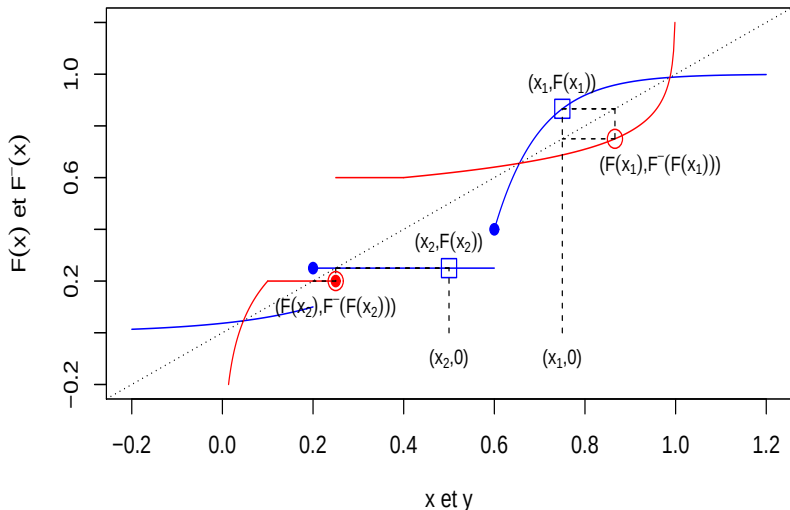
$$\begin{aligned} F^- : (0, 1) &\rightarrow \mathbb{R} \\ \alpha &\mapsto F^-(\alpha) = \inf\{x : F(x) \geq \alpha\}. \end{aligned}$$

► Illustration

Properties

- ① $\forall \alpha \in (0, 1), E_\alpha := \{x : F(x) \geq \alpha\} = [F^-(\alpha), \infty)$. ► Proof
- ② The function F^- is increasing, left continuous with right limits (whereas the cdf F is càdlàg). ► Proof

Cdf F (in blue) and F^- (in red): $F^-(F(x_1)) = x_1$ and $F^-(F(x_2)) < x_2$



Other properties

Properties

- ③ $\forall \alpha \in (0, 1), x \geq F^-(\alpha) \Leftrightarrow F(x) \geq \alpha.$
- ④ $\forall \alpha \in (0, 1), F\{F^-(\alpha)\} \geq \alpha.$
- ⑤ $\forall x \in \mathbb{R}, F^-\{F(x)\} \leq x.$
- ⑥ If $U \sim \mathcal{U}_{[0,1]}$ then $F^-(U)$ is a random variable with cdf F .
- ⑦ If F is continuous strictly increasing then $F(X) \sim \mathcal{U}_{[0,1]}.$
- ⑧ If $a \geq 0$ and $\alpha \in (0, 1), F_{aX+b}^-(\alpha) = aF_X^-(\alpha) + b.$

► Proof

Corollary (Invariance by translation and positive homogeneity)

With obvious notation, if $a \geq 0$

$$VaR_{aL+b}(\alpha) = aVaR_L(\alpha) + b.$$

A few more properties: see also [this excellent paper](#)

Properties

- ⑨ $F \circ F^- \circ F = F$
- ⑩ $F^- \circ F \circ F^- = F^-$
- ⑪ $F^-(\alpha) < x \not\Rightarrow \alpha < F(x).$
- ⑫ $\alpha < F(x) \not\Rightarrow F^-(\alpha) < x.$
- ⑬ We can have $F\{F^-(\alpha)\} > \alpha$ (iff $\alpha \notin F(\mathbb{R})$).
- ⑭ We can have $F^-\{F(x)\} < x$ (when F is constant in a neighbourhood of x).

► Proof

Interpretation of a quantile as solution of an optimization

Properties

Let $\alpha \in (0, 1)$ and the "check function" (or asymmetric absolute loss function)

► Graph

$$\rho_{\alpha}(u) = (1 - \alpha)u^{-} + \alpha u^{+} = (\alpha - 1)u1_{u < 0} + \alpha u1_{u > 0} = u(\alpha - 1_{u < 0}).$$

For all X such that $E|X| < \infty$,

$$F_X^{-}(\alpha) = \inf \left\{ \operatorname{argmin}_u E \rho_{\alpha}(X - u) \right\}.$$

► Proof

► Illustration

Remark: analogue of $EX = \operatorname{argmin}_u E(X - u)^2$ and generalization of $\operatorname{med} X = \operatorname{argmin}_u E|X - u|$.

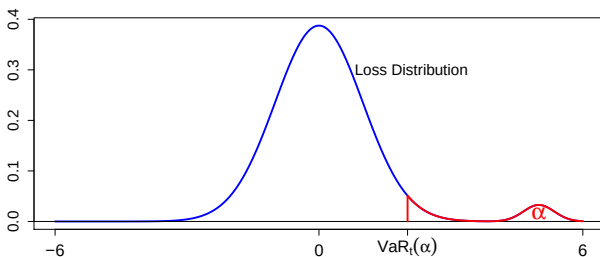
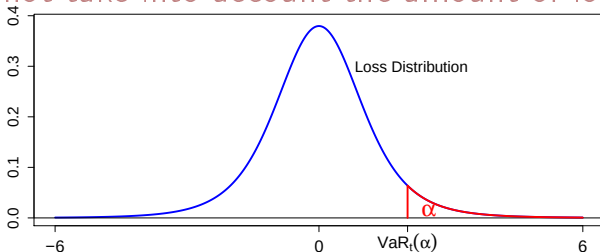
Other interpretation of the VaR

If the distribution of $L_{t,t+h}$ is continuous and integrable,

$$\begin{aligned}\text{VaR}_{t,h}(\alpha) &= \arg \min_u E \rho_{1-\alpha}(L_{t,t+h} - u) \\ &= \arg \min_u \alpha E(L_{t,t+h} - u)^- + (1 - \alpha) E(L_{t,t+h} - u)^+.\end{aligned}$$

The VaR is therefore an **optimal capital** u immobilized at the date t , causing a cost α if $L_{t,t+h} < u$ and $1 - \alpha$ if $L_{t,t+h} > u$. Usually α is small, so we give more weight to the undervaluation of capital.

VaR does not take into account the amount of losses



Definition of the Expected Shortfall

Definition (Expected Shortfall)

If $E|L_{t,t+h}| < \infty$ and $\alpha \in (0, 1)$, then

$$ES_{t,h}(\alpha) = \frac{1}{\alpha} \int_0^\alpha VaR_{t,h}(u) du.$$

The expected shortfall is the average VaR for risk levels below α . The expected shortfall is sometimes called TailVaR or AVaR (Average VaR) or even CVaR (Conditional VaR), with sometimes different definitions!

Equivalent definition

Proposition (Expected Shortfall when L is continuous)

If $L_{t,t+h}$ is continuous, strictly increasing and integrable, and if $\alpha \in (0, 1)$, then

$$\begin{aligned} ES_{t,h}(\alpha) &= E\{L_{t,t+h} \mid L_{t,t+h} \geq VaR_{t,h}(\alpha)\} \\ &= \alpha^{-1} E\left(L_{t,t+h} 1_{\{L_{t,t+h} \geq VaR_{t,h}(\alpha)\}}\right). \end{aligned}$$

► Proof

The expected shortfall is the expected loss in the event of violation of the VaR (the loss is larger than the VaR). This is the average of the $100\alpha\%$ worst losses.

Case of a Gaussian loss

Properties

If the loss $L = L_{t,h}(\alpha)$ satisfies

$$L = m + \sigma N, \quad \text{where } N \sim \mathcal{N}(0, 1),$$

and m and $\sigma > 0$ are constants, then

- ① $\text{VaR}_{t,h}(\alpha) = m + \sigma \Phi^{-1}(1 - \alpha),$
- ② $\text{ES}_{t,h}(\alpha) = m + \sigma \frac{1}{\alpha} \phi \{ \Phi^{-1}(1 - \alpha) \},$

► Proof

where ϕ and Φ denote the density and cdf of the $\mathcal{N}(0, 1)$ law.

Advantage of working with conditional loss (for the bank)

Let

$$\begin{pmatrix} Z \\ U \end{pmatrix} \sim \mathcal{N}(0, \mathbf{I}_2), \quad L = Z + U \sim \mathcal{N}(0, 2).$$

At the risk level $\alpha \in (0, 1/2)$, the VaR conditional to Z is in average smaller than the marginal VaR:

$$\begin{aligned} E\{\text{VaR}_{L|Z}(\alpha)\} &= E\{Z + \Phi^{-1}(1 - \alpha)\} = \Phi^{-1}(1 - \alpha) \\ &< \sqrt{2}\Phi^{-1}(1 - \alpha) = \text{VaR}_L(\alpha). \end{aligned}$$

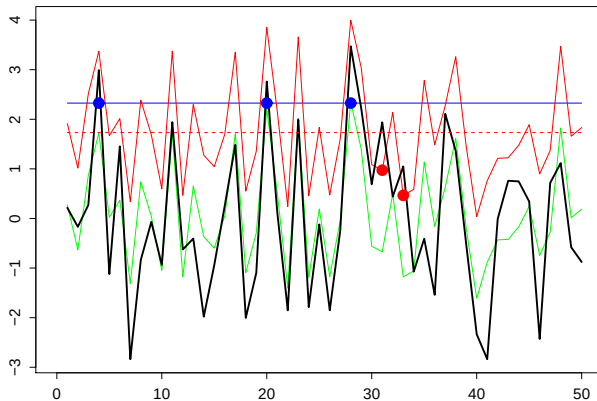
Similarly

$$E\{\text{ES}_{L|Z}(\alpha)\} < \text{ES}_L(\alpha).$$

Remark: can be extended to non centred and non Gaussian distributions; in a time series framework $L_t = E(L_t | L_u, u < t) + U_t$.

Less reserve on average with conditional VaR

Perte $L=Z+U$, information Z , VaR sachant Z et VaR marginale



The average of the conditional VaR (red dotted line) is lower than the marginal VaR (in blue)

Less reserve and also less loss

Let

$$\begin{pmatrix} Z \\ U \end{pmatrix} \sim \mathcal{N}(0, \mathbf{I}_2), \quad L = Z + U \sim \mathcal{N}(0, 2).$$

With the VaR conditional to Z , in case of a loss larger than the VaR, the **average excess loss** is

$$\begin{aligned} E\{L - \text{VaR}_{L|Z}(\alpha) \mid L > \text{VaR}_{L|Z}(\alpha)\} &= E\{U \mid U > \Phi^{-1}(1 - \alpha)\} - \Phi^{-1}(1 - \alpha) \\ &= \frac{1}{\alpha} \phi(\Phi^{-1}(1 - \alpha)) - \Phi^{-1}(1 - \alpha). \end{aligned}$$

With the marginal VaR, this excess loss is larger:

$$E\{L - \text{VaR}_L(\alpha) \mid L > \text{VaR}_L(\alpha)\} = \frac{\sqrt{2}}{\alpha} \phi(\Phi^{-1}(1 - \alpha)) - \sqrt{2} \Phi^{-1}(1 - \alpha).$$

- 1 Financial Risks
- 2 Main risk measures
- 3 Examples of methods to assess the risk of a portfolio
 - Parametric model
 - Historical Simulation
 - Riskmetrics and FHS with GARCH

Framework

Consider a crystallized (*i.e.* frozen or static) portfolio of d shares. Denote by $S_{t,j}$ the price of the stock j at time t and a_j the number of shares j . The value of the portfolio is

$$V_t = \sum_{j=1}^d a_j S_{t,j} = \mathbf{a}' \mathbf{S}_t$$

with obvious notation.

At the horizon h the loss is

$$L_{t,t+h} = -(V_{t+h} - V_t) = -\mathbf{a}' \sum_{j=1}^h \Delta \mathbf{S}_{t+j}, \quad \Delta \mathbf{S}_t = \mathbf{S}_t - \mathbf{S}_{t-1}.$$

Marginal VaR with static parametric model

If, for instance, we assume $\Delta \mathbf{S}_t \sim \mathcal{N}(\mathbf{m}, \Sigma)$, then

$$\text{VaR}_{t,1}(\alpha) = -\mathbf{a}'\mathbf{m} + \sqrt{\mathbf{a}'\Sigma\mathbf{a}}\Phi^{-1}(1-\alpha).$$

This is an unconditional/marginal VaR.

In practice $\mathbf{m}$ and Σ are estimated by the empirical mean and variance of $\Delta \mathbf{S}_i, i = 2, \dots, t$.

Assuming $\Delta \mathbf{S}_t$ iid $\mathcal{N}(\mathbf{m}, \Sigma)$ we have

$$L_{t,t+h} \sim \mathcal{N}(-h\mathbf{a}'\mathbf{m}, h\mathbf{a}'\Sigma\mathbf{a}),$$

thus

$$\text{VaR}_{t,h}(\alpha) = -h\mathbf{a}'\mathbf{m} + \sqrt{h}\sqrt{\mathbf{a}'\Sigma\mathbf{a}}\Phi^{-1}(1-\alpha),$$

and, when $\mathbf{m} = 0$,

$$\text{VaR}_{t,h}(\alpha) = \sqrt{h}\text{VaR}_{t,1}(\alpha).$$

In practice, the Gaussian iid model seems unrealistic.

Historical Simulation: Marginal VaR with non parametric model

The distribution of $\Delta \mathbf{S}_{t+j}$ is estimated by the empirical distribution of $\Delta \mathbf{S}_i, i=2, \dots, t$. Thus $\text{VaR}_{t,1}(\alpha)$ is estimated by the opposite of the empirical α -quantile of $\mathbf{a}'\Delta \mathbf{S}_i, i=2, \dots, t$.

For more general loss functions

$$L_{t,t+h} = \ell_t(\Delta \mathbf{S}_{t+1}, \dots, \Delta \mathbf{S}_{t+h}),$$

one can simulate $L_{t,t+h}$ (generate "scenarios" for the future loss) by sampling $\Delta \mathbf{S}_{t+1}, \dots, \Delta \mathbf{S}_{t+h}$ in $\Delta \mathbf{S}_i, i=2, \dots, t$.

To make "time-varying" this VaR, the method can be used on "rolling windows". For example, one can generate scenarios by sampling in $\Delta \mathbf{S}_i, i=t-K+1, \dots, t$ (with for instance $K=250$). The use of rolling windows, however, lacks a theoretical foundation.

Riskmetrics: conditional VaR by exponential smoothing (JP Morgan)

Let the return $r_t = \mathbf{a}' \Delta \mathbf{S}_t$. The "model" is

$$L_{t,t+1} = -r_{t+1} = \sigma_{t+1} \eta_{t+1}, \quad \eta_t \text{ iid } \mathcal{N}(0, 1)$$

where

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) r_t^2$$

with $\lambda \in (0, 1)$ a smoothing parameter ($\lambda = 0.95$ in general).

Starting from an initial value σ_1^2 , we compute

$$\sigma_j^2 = \lambda \sigma_{j-1}^2 + (1 - \lambda) r_{j-1}^2, \quad j = 2, \dots, t+1$$

and we obtain

$$\text{VaR}_{t,1}(\alpha) = \sigma_{t+1} \Phi^{-1}(1 - \alpha).$$

This method also lacks a theoretical foundation.

Semi-parametric Filtered Historical Simulation method

Consider the GARCH model

$$L_{t,t+1} = -r_{t+1} = \sigma_{t+1}\eta_{t+1}, \quad \eta_t \text{ iid}(0, 1),$$

where

$$\sigma_{t+1}^2 = \omega + \alpha r_t^2 + \beta \sigma_t^2.$$

Starting from an initial value σ_1^2 and using an estimator of (ω, α, β) , we compute

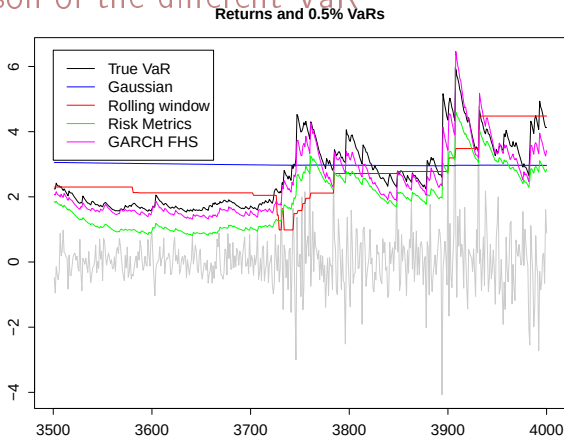
$$\hat{\sigma}_{j+1}^2 = \hat{\omega} + \hat{\alpha} r_j^2 + \hat{\beta} \hat{\sigma}_j^2, \quad \hat{\eta}_j = \frac{r_j}{\hat{\sigma}_j}, \quad j = 1, \dots, t,$$

and set

$$\text{VaR}_{t,1}(\alpha) = \hat{\sigma}_{t+1} \hat{\xi}_{1-\alpha}$$

where $\hat{\xi}_{1-\alpha}$ is the empirical $1 - \alpha$ quantile of $\hat{\eta}_j$ for $j = 1, \dots, t$.

Comparison of the different VaR



DGP: $\epsilon_t = \sigma_t \eta_t$, $\eta_t \sim \text{St}_v$, $\sigma_t^2 = \omega + \alpha + \epsilon_{t-1}^2 1_{\epsilon_{t-1} > 0} + \alpha - \epsilon_{t-1}^2 1_{\epsilon_{t-1} < 0} + \beta \sigma_{t-1}^2$

Fin du Chapitre 1 😊 !

Proof that $E_\alpha := \{x: F(x) \geq \alpha\} = [F^-(\alpha), \infty)$.

For $\alpha \in (0, 1)$, the set E_α is not empty, otherwise $F(x) < \alpha < 1 \ \forall x$, which is impossible because $F(x) \rightarrow 1$ as $x \rightarrow \infty$. Note that $x \in E_\alpha$ and $x' \geq x$ imply $F(x') \geq F(x) \geq \alpha$, and thus $x' \in E_\alpha$. We thus have either $E_\alpha = (a, \infty)$ or $E_\alpha = [a, \infty)$, with $a \in \mathbb{R} \cup \{-\infty\}$. The set E_α has a lower bound (i.e. $a \neq -\infty$), otherwise $F(x) \geq \alpha > 0 \ \forall x$, which is impossible since $F(x) \rightarrow 0$ as $x \rightarrow -\infty$. Now note that if $x_n \downarrow x$ and $x_n \in E_\alpha \ \forall n$ then $x \in E_\alpha$ since F is right continuous. We thus have shown

$$E_\alpha = [F^-(\alpha), \infty[.$$

◀ Properties of F^-

Proof of Property 2

Let $0 < \alpha \leq \alpha' < 1$. If $x \in E_{\alpha'}$ then $F(x) \geq \alpha' \geq \alpha$, that is $x \in E_{\alpha}$. Using Property 1, we have $E_{\alpha'} = [F^{-}(\alpha'), \infty) \subset E_{\alpha} = [F^{-}(\alpha), \infty)$, which implies $F^{-}(\alpha) \leq F^{-}(\alpha')$ and shows that F^{-} is increasing.

Let α_n increasing to $\alpha \in (0, 1)$. Note that $x \in [F^{-}(\alpha_n), F^{-}(\alpha))$ iff $\alpha_n \leq F(x) < \alpha$. This can not be true for all n , since $\alpha_n \rightarrow \alpha$. It follows that $F^{-}(\alpha_n) \rightarrow F^{-}(\alpha)$, hence the left continuity.

Let the sequence α_n which decreases to $\alpha \in (0, 1)$. The sequence $F^{-}(\alpha_n)$ is decreasing and bounded below by $F^{-}(\alpha)$, and thus it converges. This shows that F^{-} admits a right limit at any point.

◀ Properties of F^{-}

Proof of the other properties

- ③ By definition of E_α , $F(x) \geq \alpha \Leftrightarrow x \in E_\alpha$, and in view of Property 1, $x \in E_\alpha \Leftrightarrow x \geq F^-(\alpha)$.
- ④ By Property 3, $F\{F^-(\alpha)\} \geq \alpha$ iff $F^-(\alpha) \geq F^-(\alpha)$, which is obviously true.
- ⑤ By Property 3, $F^-\{F(x)\} \leq x$ iff $F(x) \leq F(x)$.
- ⑥ Again by Property 3, we have $P\{F^-(U) \leq x\} = P\{U \leq F(x)\}$. Since the cdf of U is the identity on $[0, 1]$, we have $P\{U \leq F(x)\} = F(x)$, and thus $P\{F^-(U) \leq x\} = F(x)$.
- ⑦ $P(F(X) \leq \alpha) = P(X \leq F^{-1}(\alpha)) = F(F^{-1}(\alpha)) = \alpha$.
- ⑧ For $a > 0$, $F_{aX+b}^-(\alpha) \leq x \Leftrightarrow \alpha \leq F_{aX+b}(x) = P(X \leq \frac{x-b}{a}) \Leftrightarrow F_X^-(\alpha) \leq \frac{x-b}{a} \Leftrightarrow aF_X^-(\alpha) + b \leq x, \forall x$.

◀ Properties of F^-

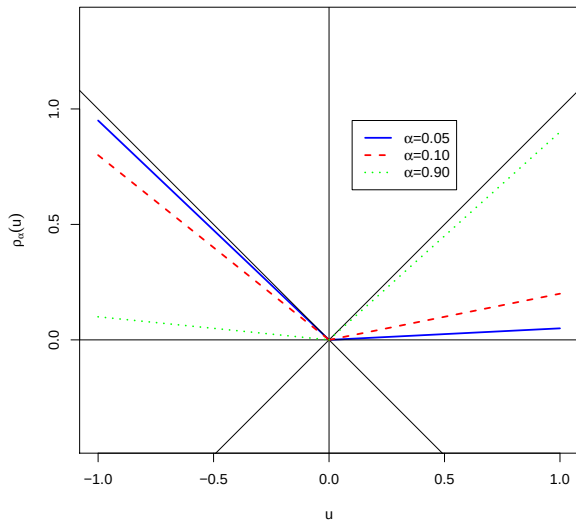
More properties (continued)

- 9 By Property 5 and the fact that F is increasing we have $F(F^-(F(x))) \leq F(x)$. By Property 4, with $\alpha = F(x)$, we also have

$$F(F^-(F(x))) = F(F^-(\alpha)) \geq \alpha = F(x).$$

- 10 By Property 5 with $x = F^-(\alpha)$, we have $F^-(F(F^-(\alpha))) \leq F^-(\alpha)$. By Property 4, we have $F(F^-(\alpha)) \geq \alpha$, and since F^- is not decreasing we also have $F^-(F(F^-(\alpha))) \geq F^-(\alpha)$. ◀ Properties of F^-

Graph of the Check Function



Proof that the quantile is solution of the minimizing problem

- The function $\rho_\alpha(\cdot)$ is convex. It follows that $u \mapsto \rho_\alpha(X - u)$ is a.s. convex on $\mathbb{R}$. By sum of convex functions, $u \mapsto E\rho_\alpha(X - u)$ is also convex on $\mathbb{R}$ (and thus continuous).
- Fatou's lemma entails

$$\liminf_{|u| \rightarrow \infty} E\rho_\alpha(X - u) \geq E \liminf_{|u| \rightarrow \infty} \rho_\alpha(X - u) = +\infty.$$

- Therefore $u \mapsto E\rho_\alpha(X - u)$ admits a minimum, at a single point or on a finite interval. Let $u^* = \inf\{\operatorname{argmin}_u E\rho_\alpha(X - u)\}$. The derivative at u^* does not always exist, but the left derivative must be strictly negative before u^* and the right derivative must be ≥ 0 after u^* :

► Illustration

$$u^* = \inf \left\{ u : \frac{\partial}{\partial u^+} E\rho_\alpha(X - u) \geq 0 \right\} = \sup \left\{ u : \frac{\partial}{\partial u^-} E\rho_\alpha(X - u) < 0 \right\}.$$

Proof (continued)

- The (right and left) derivative of

$$u \mapsto \rho_\alpha(x-u) = (\alpha-1)(x-u)1_{x-u \leq 0} + \alpha(x-u)1_{x-u > 0}$$

is equal to $1-\alpha$ when $u > x$ and $-\alpha$ when $u < x$. At $u = x$, the left derivative is $-\alpha$ and the right derivative is $1-\alpha$. We thus have

$$\frac{\partial}{\partial u^-} \rho_\alpha(x-u) = -\alpha 1_{u \leq x} + (1-\alpha) 1_{u > x} = -\alpha + 1_{u > x},$$

$$\frac{\partial}{\partial u^+} \rho_\alpha(x-u) = -\alpha 1_{u < x} + (1-\alpha) 1_{u \geq x} = -\alpha + 1_{u \geq x}.$$

- Using the dominated convergence theorem to derive under the expectation,

$$\frac{\partial}{\partial u^+} E \rho_\alpha(X-u) = -\alpha + P(X \leq u),$$

therefore u^* is the α -quantile.

Proof of the equivalence of the two definitions of the ES

Let $L \sim L_{t,t+h}$ at time t and $U \sim \mathcal{U}_{[0,1]}$. We know that $F_L^{-1}(U) \sim L$. We thus have

$$\begin{aligned} E\{L 1_{L > \text{VaR}_{t,h}(\alpha)}\} &= E\{F_L^{-1}(U) 1_{F_L^{-1}(U) > F_L^{-1}(1-\alpha)}\} \\ &= E\{F_L^{-1}(U) 1_{U > 1-\alpha}\} \\ &= \int_{1-\alpha}^1 F_L^{-1}(u) du \\ &= \int_0^\alpha \text{VaR}_{t,h}(v) dv \end{aligned}$$

by the change of variable $v = 1 - u$.

[◀ Return](#)

Proof of the expressions for a Gaussian loss

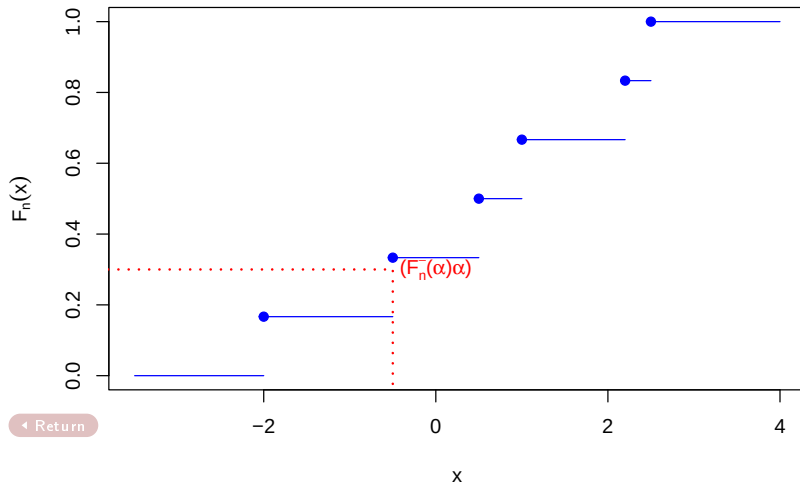
The expression of the VaR comes from $F_{aX+b}^-(\alpha) = b + aF_X^-(\alpha)$ for $a \geq 0$. By the change of variable $v = \Phi^{-1}(1-u)$, we have

$$\begin{aligned} \text{ES}_{t,h}(\alpha) &= m + \sigma \frac{1}{\alpha} \int_0^\alpha \Phi^{-1}(1-u) du \\ &= m + \sigma \frac{1}{\alpha} \int_{\Phi^{-1}(1-\alpha)}^\infty v \phi(v) dv. \end{aligned}$$

Alternative proof:

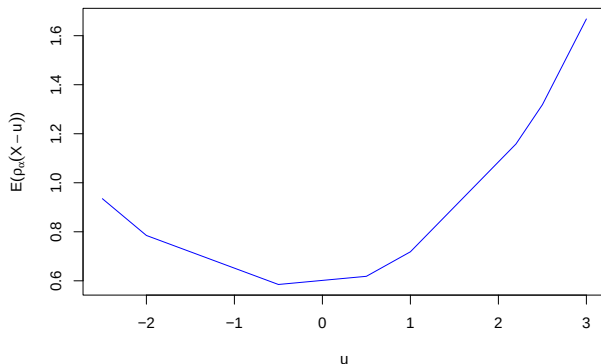
$$\begin{aligned} E(L | L > \text{VaR}) &= m + \sigma E\{N | N > \Phi^{-1}(1-\alpha)\} \\ &= m + \sigma \frac{1}{\alpha} \int_{\Phi^{-1}(1-\alpha)}^\infty u \phi(u) du. \end{aligned}$$

X uniformly distributed on $-2, -0.5, 0.5, 1, 2.2, 2.5$; $\alpha = 0.3$

[Return](#)

X uniformly distributed on $-2, -0.5, 0.5, 1, 2.2, 2.5$

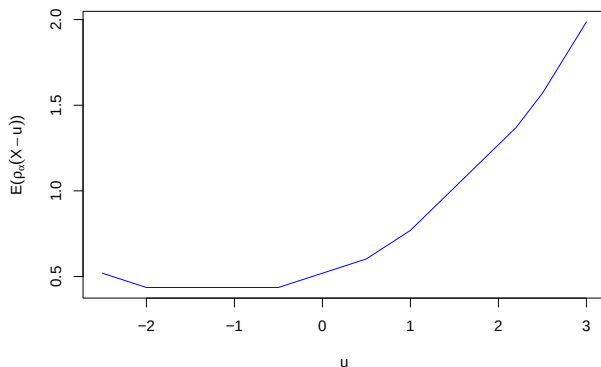
$\alpha=0.3$



◀ Return

X uniformly distributed on $-2, -0.5, 0.5, 1, 2.2, 2.5$

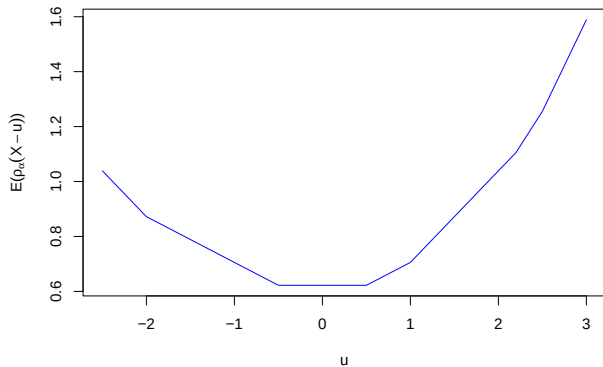
$\alpha=1/6$



◀ Return

X uniformly distributed on $-2, -0.5, 0.5, 1, 2.2, 2.5$

$\alpha=1/3$



[Return](#)

$u^* = -0.5$, $\frac{\partial}{\partial u} E\rho_\alpha(X-u) < 0$ for $u < u^*$, $\frac{\partial}{\partial u^+} E\rho_\alpha(X-u) \geq 0$ for $u > u^*$